

ANSWER KEY

GATE CS — GATE CS 2026 Trend-Based Mock Paper
2

Q.No.	Type	Answer	Marks
1	2MSQ	A	2
2	2NAT	B	2
3	2NAT	Hive	2
4	2NAT	" (ab)	2
5	2MCQ	C	2
6	2MCQ	C	2
7	2NAT	A	2
8	2NAT	1800	2
9	2MCQ	B	2
10	2MCQ	C	2
11	2MCQ	C	2
12	2NAT	B	2
13	2NAT	B	2
14	2MCQ	B	2
15	2MSQ	C	2
16	2NAT	A	2
17	1NAT	B	1
18	1MCQ	C	1
19	1NAT	B	1
20	1MCQ	C	1
21	1MCQ	B	1
22	2MCQ	C	2
23	1MCQ	B	1
24	1NAT	C	1
25	2NAT	C	2
26	1MCQ	C	1
27	1MCQ	B	1
28	1MCQ	A	1
29	1NAT	B	1
30	1MCQ	A	1
31	1MCQ	B	1
32	1MCQ	C	1
33	1NAT	B	1
34	1MCQ	B	1
35	1NAT	B	1
36	1MCQ	B	1

37	1NAT	C	1
38	1MCQ	A	1
39	1NAT	B	1
40	1MCQ	B	1
41	1MCQ	B	1
42	1MCQ	B	1
43	1MCQ	B	1
44	1MCQ	A	1
45	2MCQ	6	2
46	1MCQ	A	1
47	2MCQ	C. 1% loss	2
48	1MCQ	A	1
49	2MSQ	Insufficient information	2
50	2MCQ	14	2
51	2MCQ	A	2
52	2NAT	3	2
53	2MCQ	D. 62.5%	2
54	2MCQ	A	2
55	2MCQ	A	2
56	1MCQ	resolve	1
57	2NAT	120	2
58	2MCQ	B	2
59	2NAT	B	2
60	2NAT	2	2
61	1MCQ	A	1
62	2MCQ	A	2
63	2NAT	A	2
64	2NAT	3	2
65	2NAT	B	2

Q1. General Aptitude | 2 marks | 2MSQ

Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select ...

Answer: A

The sentence requires indefinite articles: "Ravi had a younger brother..." (a younger brother). "...who taught at a university" (a university, since university begins with a consonant sound). "...as an honorable man" (an, because honorable starts with a vowel sound). Hence the correct sequence is a; a; an, which corresponds to option A.

Source: Adapted from GATE 2025 CS Session 1 Q1

Q2. General Aptitude | 2 marks | 2NAT

Antonym of 'ephemeral':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q10

Q3. General Aptitude | 2 marks | 2NAT

Bird : Nest :: Bee : _____ Select the correct option to complete the analogy.

Answer: Hive

A bird lives in a nest that it builds, just as a bee lives in a hive that it constructs; the relationship is analogous.

Source: Adapted from GATE 2025 CS Session 2 Q2

Q4. General Aptitude | 2 marks | 2NAT

For positive non-zero real variables and , if $\ln a + \ln b = \ln(a+b)$ then, the value of $\frac{a+b}{ab}$ is

Answer: " (ab)

Given $\ln(a+b) = \ln a + \ln b = \ln(ab)$, exponentiating both sides gives $a+b = (ab)$.

Source: Adapted from GATE 2024 CS Session 2 Q4

Q5. General Aptitude | 2 marks | 2MCQ

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the...

Answer: C

The sentence contrasts initial hesitation with a steadfast attitude. 'Resolve' means firm determination, fitting the idea that his willingness never wavered, whereas the other words do not convey this meaning.

Source: Adapted from GATE 2025 CS Session 2 Q1

Q6. General Aptitude | 2 marks | 2MCQ

If 'f' denotes increasing order of intensity, then the meaning of the words [walk jog sprint] is analogous to [bothered ...]

Answer: C

The sequence walk < jog < sprint shows increasing intensity. We need a word of intermediate intensity between 'bothered' and 'daunted'. Among the options, only 'fazed' (C) denotes a mild disturbance, fitting between being bothered and being daunted.

Source: Adapted from GATE 2024 CS Session 2 Q1

Q7. General Aptitude | 2 marks | 2NAT

In a class of 60 students, 30 play cricket, 25 play football, and 15 play both. How many play neither?

Answer: A

Correct Answer: A

Source: Adapted from GATE 2024 CS Session 1 Q10

Q8. General Aptitude | 2 marks | 2NAT

In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of s...

Answer: 1800

Let C be students who like core branches, O those who like other branches. Given $C = \frac{1}{4} O$, so $O = 4C$. Both like core and other = 500. Students who like at least one = total - neither = $10000 - 1500 = 8500$. Using inclusion - exclusion: $C + O - 500 = 8500$. Substitute $O = 4C$: $C + 4C - 500 = 8500 \Rightarrow 5C = 9000 \Rightarrow C = 1800$.
Source: Adapted from GATE 2024 CS Session 2 Q3

Q9. Engineering Mathematics | 2 marks | 2MCQ

Determinant of 2x2 identity:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q1

Q10. Engineering Mathematics | 2 marks | 2MCQ

Probability of at least one head in 3 coin tosses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q11

Q11. Engineering Mathematics | 2 marks | 2MCQ

Let A be a 3x3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q1

Q12. Engineering Mathematics | 2 marks | 2NAT

Mean of binomial $n=10, p=0.5$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q1

Q13. Engineering Mathematics | 2 marks | 2NAT

Derivative of x^3 at $x=2$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q1

Q14. Programming and Data Structures | 2 marks | 2MCQ

Infix to postfix uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q6

Q15. Programming and Data Structures | 2 marks | 2MSQ

Avg hash lookup:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q3

Q16. Programming and Data Structures | 2 marks | 2NAT
Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q17. Programming and Data Structures | 1 marks | 1NAT
What is the time complexity of inserting at the beginning of a linked list?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q3

Q18. Algorithms | 1 marks | 1MCQ
Kruskal uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q2

Q19. Algorithms | 1 marks | 1NAT
Worst-case time complexity of merge sort:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q12

Q20. Algorithms | 1 marks | 1MCQ
LCS uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 CS Session 1 Q7

Q21. Algorithms | 1 marks | 1MCQ
Time complexity of binary search:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q11

Q22. Algorithms | 2 marks | 2MCQ
The time complexity of Dijkstra's algorithm using a binary heap is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q7

Q23. Algorithms | 1 marks | 1MCQ
BFS uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q7

Q24. Algorithms | 1 marks | 1NAT
Merge sort space:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q7

Q25. Operating Systems | 2 marks | 2NAT
*Using 3 frames, which page replacement algorithm would not cause any page fault for reference string
0,1,2,3,0,1,4,0,1,2...*

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q8

Q26. Operating Systems | 1 marks | 1MCQ
Which of the following is NOT a necessary condition for deadlock?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q4

Q27. Operating Systems | 1 marks | 1MCQ
Semaphore with value 1 is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q8

Q28. Operating Systems | 1 marks | 1MCQ
TLB stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 CS Session 1 Q9

Q29. Operating Systems | 1 marks | 1NAT
Banker's algorithm is for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q4

Q30. Operating Systems | 1 marks | 1MCQ
Round-robin uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q4

Q31. Operating Systems | 1 marks | 1MCQ
Inode contains:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q4

Q32. Computer Networks | 1 marks | 1MCQ
Which protocol operates at the application layer of the TCP/IP model?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q6

Q33. Computer Networks | 1 marks | 1NAT
CSMA/CD is used in:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q13

Q34. Computer Networks | 1 marks | 1MCQ
OSPF is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q6

Q35. Computer Networks | 1 marks | 1NAT
Protocol to find MAC from IP:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q6

Q36. Computer Networks | 1 marks | 1MCQ
TCP connection establishment uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q14

Q37. Computer Networks | 1 marks | 1NAT
Asymmetric encryption:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q5

Q38. Digital Logic | 1 marks | 1MCQ
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q39. Digital Logic | 1 marks | 1NAT
Full adder has how many inputs:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q2

Q40. Digital Logic | 1 marks | 1MCQ
The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q41. Theory of Computation | 1 marks | 1MCQ
Which language is regular?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q13

Q42. Theory of Computation | 1 marks | 1MCQ
PDA recognizes:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q14

Q43. Theory of Computation | 1 marks | 1MCQ
Halting problem:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q8

Q44. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q45. CS | 2 marks | 2MCQ
Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a ...

Answer: 6

The original 1400 byte payload is first fragmented on the 542 byte MTU link into three pieces of 520, 520 and 360 bytes of data. On the second link (MTU 360), each of these fragments is further split into sub fragments with a maximum data size of 336 bytes (multiple of 8). Thus 520 !' 336+184 (2 fragments), another 520 !' 336+184 (2 fragments), and 360 !' 336+24 (2 fragments). Total fragments received = 2+2+2 = 6.

Source: Adapted from GATE 2024 CS Session 1 Q65

Q46. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q47. CS | 2 marks | 2MCQ
A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In ...

Answer: C. 1% loss

Let the common selling price be S . For a 10% profit, cost = $S/1.10$; for a 10% loss, cost = $S/0.90$. Total cost = $S/1.10 + S/0.90 = 2.0202S$. Total selling = $2S$. Loss = $2S - 2.0202S = -0.0202S$. Loss % = $(0.0202S / 2.0202S) * 100$ "H 1%.

Hence overall a 1% loss.

Source: Adapted from GATE 2024 CS Session 2 Q7

Q48. Digital Logic | 1 marks | 1MCQ
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q49. CS | 2 marks | 2MSQ
Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial s...

Answer: Insufficient information

The question refers to a specific sequential circuit diagram that is not provided. Without knowing the number of flip-flops and the interconnections, the length of the state cycle cannot be determined.

Source: Adapted from GATE 2025 CS Session 1 Q60

Q50. CS | 2 marks | 2MCQ
Consider the following code snippet using the fork() and wait() system calls. Assume that the code compiles and runs correctly...

Answer: 14

Each loop iteration doubles the number of processes. Iteration 1: 2 processes print, iteration 2: 4 processes print, iteration 3: 8 processes print. Total prints = $2 + 4 + 8 = 14$.

Source: Adapted from GATE 2024 CS Session 1 Q57

Q51. CS | 2 marks | 2MCQ
Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation R1+R2, which one of the...

Answer: A

$1011 (-5) + 1110 (-2) = 10011 (-7)$ in 4-bit 2's complement. No overflow since signs are different. Other pairs overflow.

Source: Adapted from GATE 2022 CS Session 1 Q18

Q52. CS | 2 marks | 2NAT
Consider the following C program: #include <stdio.h> int gate(int n) { int d, t, newnum, turn; newnum = turn = 0; t=1; ...

Answer: 3

The function counts digits. gate(100) processes '100' - 3 digits, so returns 3.

Source: Adapted from GATE 2025 CS Session 1 Q63

Q53. CS | 2 marks | 2MCQ
The pie charts depict the shares of various power generation technologies in the total electricity generation of a country...

Answer: D. 62.5%

Renewable share 2007 = Hydro 30% + Solar 5% + Wind 5% = 40%. Renewable share 2023 = Hydro 35% + Solar 20% + Wind 10% = 65%. Increase = $65\% - 40\% = 25$ percentage points. Percentage increase = $(25/40) \times 100 = 62.5\%$.

Source: Adapted from GATE 2024 CS Session 2 Q8

Q54. CS | 2 marks | 2MCQ
Suppose we are given n keys, m hash table slots, and two simple uniform hash functions h1 and h2. Further suppose our h...

Answer: A

Double hashing load factor; Expected successful search cost is $(1/\alpha) \ln(1/\alpha)$.

Source: Adapted from GATE 2022 CS Session 1 Q16

Q55. CS | 2 marks | 2MCQ
Consider a process P running on a CPU. Which one or more of the following events will always trigger a context switch by...

Answer: A

Blocking system call ALWAYS triggers context switch. Page fault may not always (OS-dependent). Timer interrupt may not always (if same process resumes).

Source: Adapted from GATE 2024 CS Session 2 Q25

Q56. General Aptitude | 1 marks | 1MCQ

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the...

Answer: resolve

The sentence contrasts initial hesitation with unwavering willingness; 'resolve' means firm determination, fitting the context, whereas the other options do not convey this meaning.

Source: Adapted from GATE 2025 CS Session 2 Q1

Q57. CS | 2 marks | 2NAT

$S = \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let f be the set of functions from S to itself. For any two functi...

Answer: 120

Functions from $\{0,1,2,3\}$ to itself: total $4^4 = 256$. Counting functions satisfying given property: 120.

Source: Adapted from GATE 2025 CS Session 1 Q49

Q58. CS | 2 marks | 2MCQ

The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's alg...

Answer: B

Booth's algorithm for 4-bit signed multiplication. After processing all bits, result matches option B.

Source: Adapted from GATE 2025 CS Session 2 Q32

Q59. CS | 2 marks | 2NAT

Sequence the following sentences in a coherent passage. P: This fortuitous geological event generated a colossal amount ...

Answer: B

Logical sequence: Geological event !' energy release !' dinosaur extinction !' fossil formation !' discovery by paleontologists.

Source: Adapted from GATE 2024 CS Session 2 Q6

Q60. CS | 2 marks | 2NAT

The value of the following limit is _____. $\lim_{x \rightarrow 0} \frac{1 - e^{-2x}}{x^2}$

Answer: 2

$\lim_{x \rightarrow 0} \frac{(1 - e^{-2x})}{x^2} = 2$ using L'Hôpital's rule or Taylor series expansion.

Source: Adapted from GATE 2022 CS Session 1 Q34

Q61. General Aptitude | 1 marks | 1MCQ

If $f(x) = -f(-x)$ for all real values of x , which one of the following statements is true?

Answer: A

The given condition implies $f(x) = -f(-x)$ for every real x . Adding $f(x)$ to both sides gives $2f(x)=0$, so $f(x)=0$ for all x . Consequently $f(-x)=0$ as well. Hence both $f(x)$ and $f(-x)$ are zero, matching option A.

Source: Adapted from GATE 2025 CS Session 2 Q3

Q62. CS | 2 marks | 2MCQ

Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant is added to the weight ...

Answer: A

Adding constant to all edges preserves edge weight differences. MSTs and shortest paths remain unchanged.

Source: Adapted from GATE 2025 CS Session 2 Q37

Q63. CS | 2 marks | 2NAT

Consider the following recurrence relation: $T(n) = T(n-1) + 1$, $T(1) = 1$. Which one of the following options is CORRECT?

Answer: A

$T(n) = T(n-1) + 1$ with $T(1) = 1$. Solution: $T(n) = n(n+1)/2 = 9, \dots$

Source: Adapted from GATE 2024 CS Session 1 Q42

Q64. CS | 2 marks | 2NAT

Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. T...

Answer: 3

Extract-Max from max-heap of 8 elements requires $\log_2(8) = 3$ comparisons in worst case.

Source: Adapted from GATE 2023 CS Session 1 Q46

Q65. CS | 2 marks | 2NAT

Consider the operator precedence and associativity rules for the integer arithmetic operators given in the table below. ...

Answer: B

* and / have higher precedence than + and -. Same precedence operators are left-associative. Expression evaluates to specific value.

Source: Adapted from GATE 2024 CS Session 1 Q33